STATISTICS · STATISTICAL INFERENCE

Significance Test & Power Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Significance Test & Power Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A coin lands heads 60 times in 100 flips. If the coin is fair, how unusual is 60 or more heads? Estimate.

2. If H_0 is rejected at $\alpha = 0.05$, must it also be rejected at $\alpha = 0.01$? Explain.

2 Explore use the tool

1. H_0: $p = 0.5$, H_a: $p > 0.5$, $n = 100$. Find the smallest number of successes that is significant at $\alpha = 0.05$. Record z and the p-value.

2. With 59 successes, switch to H_a: $p \neq 0.5$. Record the p-value. Is the result still significant?

3. Keep $\hat{p} = 0.55$. Record the one-sided p-values for n = 100, 200 and 400.

4. Errors & power: $p_0 = 0.5$, H_a: $p > 0.5$, $\alpha = 0.05$, true $p = 0.6$. Find the smallest n on the slider with power ≥ 0.80.

3 Record use the tool and fill in the table

Power (H_0: $p = 0.5$, H_a: $p > 0.5$)

Use the Errors & power tab.

n	α	True p	Power	β = P(Type II error)
100	0.05	0.60		
150	0.05	0.60		
100	0.01	0.60		
100	0.05	0.55		
100	0.05	0.50		

4 Explain what did you discover?

1. The p-value is the probability, assuming _____ is true, of a result at least as _____ as the one observed.
2. If the p-value ≤ α, we _____ H_0. Otherwise we _____ H_0.
3. A Type I error is _____. A Type II error is _____.
4. Power increases when n _____, when α _____, or when the true value is _____ p_0.
5. A study reports a p-value of 0.30 and concludes, "The null hypothesis is true." What is wrong with this conclusion?

5 Apply a new situation, no tool

1. H_0: $p = 0.30$, H_a: $p > 0.30$. An SRS of 200 has 72 successes. Find z and the p-value, and state a conclusion at $\alpha = 0.05$.

2. H_0: a new drug is no better than the old one. Describe a Type I error and a Type II error in context, and a consequence of each.

3. How does changing α from 0.01 to 0.05 affect the probability of a Type I error and the power?